[2]:
```python
# Train Seat Reservation System

seats = [0] * 20

while True:
    print("\n1.Book Seat")
    print("2.Cancel Seat")
    print("3.Show Seats")
    print("4.Booking Percentage")
    print("5.Exit")

    choice = int(input("Enter choice: "))

    if choice == 1:
        seat_no = int(input("Enter seat number (1-20): "))

        if seats[seat_no - 1] == 0:
            seats[seat_no - 1] = 1
            print("Seat booked successfully.")
        else:
            print("Seat already booked.")

    elif choice == 2:
        seat_no = int(input("Enter seat number to cancel: "))

        if seats[seat_no - 1] == 1:
            seats[seat_no - 1] = 0
            print("Seat cancelled successfully.")
```

```
        else:
            print("Seat is already available.")

    elif choice == 3:
        print("\nSeat Status:")
        for i in range(20):
            print("Seat", i + 1, ":", seats[i])

    elif choice == 4:
        booked = seats.count(1)
        percentage = (booked / 20) * 100
        print("Booking Percentage =", percentage, "%")

    elif choice == 5:
        print("Exiting...")
        break

    else:
        print("Invalid Choice")
```

```
1.Book Seat
2.Cancel Seat
3.Show Seats
4.Booking Percentage
5.Exit
Enter choice:  4
Booking Percentage = 0.0 %
```

```
1.Book Seat
2.Cancel Seat
3.Show Seats
4.Booking Percentage
5.Exit
Enter choice:  5
Exiting...
```

[3]:

```python
# YouTube Playlist Analyzer

videos = [12, 4, 8, 15, 3, 20]

total_time = sum(videos)
longest_video = max(videos)
average_duration = total_time / len(videos)

print("Total Watch Time =", total_time, "minutes")
print("Longest Video =", longest_video, "minutes")
print("Average Video Duration =", average_duration)

print("Videos shorter than 5 minutes:")
for v in videos:
    if v < 5:
        print(v)
```

```
Total Watch Time = 62 minutes
Longest Video = 20 minutes
Average Video Duration = 10.333333333333334
Videos shorter than 5 minutes:
4
3
```

[4]:

```python
# Social Media Hashtag Tracker

hashtags = ["#fun", "#python", "#fun", "#code", "#python"]

frequency = {}

for tag in hashtags:
    if tag in frequency:
        frequency[tag] += 1
    else:
        frequency[tag] = 1

# Most repeated hashtag
max_count = 0
most_repeated = ""

for tag in frequency:
    if frequency[tag] > max_count:
        max_count = frequency[tag]
        most_repeated = tag

print("Most Repeated Hashtag:", most_repeated)

# Unique hashtags
unique_tags = list(frequency.keys())
print("Unique Hashtags:", unique_tags)

# Frequency of each hashtag
print("Frequency of each hashtag:")
```

```python
for tag in frequency:
    print(tag, ":", frequency[tag])

# Remove duplicates
no_duplicates = []

for tag in hashtags:
    if tag not in no_duplicates:
        no_duplicates.append(tag)

print("After Removing Duplicates:", no_duplicates)
```

```
Most Repeated Hashtag: #fun
Unique Hashtags: ['#fun', '#python', '#code']
Frequency of each hashtag:
#fun : 2
#python : 2
#code : 1
After Removing Duplicates: ['#fun', '#python', '#code']
```

[5]:

```python
# Expense Splitter

expenses = [1200, 850, 1500, 950]

total_expense = sum(expenses)
average_share = total_expense / len(expenses)

highest_paid = max(expenses)
person = expenses.index(highest_paid) + 1
```

```python
print("Total Expense =", total_expense)
print("Average Share =", average_share)
print("Person", person, "paid the highest amount =", highest_paid)

print("\nEqual Contribution Details:")

for i in range(len(expenses)):
    difference = expenses[i] - average_share

    if difference > 0:
        print("Person", i + 1, "should receive", difference)
    elif difference < 0:
        print("Person", i + 1, "should pay", abs(difference))
    else:
        print("Person", i + 1, "paid exact share")
```

```
Total Expense = 4500
Average Share = 1125.0
Person 3 paid the highest amount = 1500

Equal Contribution Details:
Person 1 should receive 75.0
Person 2 should pay 275.0
Person 3 should receive 375.0
Person 4 should pay 175.0
```

[6]:

```python
# Pattern Compression

numbers = [1, 1, 1, 2, 2, 3, 3, 3]

compressed = []

count = 1

for i in range(len(numbers)):

    if i < len(numbers) - 1 and numbers[i] == numbers[i + 1]:
        count += 1
    else:
        compressed.append((numbers[i], count))
        count = 1

print("Compressed List:")
print(compressed)
```

```
Compressed List:
[(1, 3), (2, 2), (3, 3)]
```

```
[7]: # Mini Shopping Cart System

cart = []
prices = []

while True:
    print("\n1.Add Item")
    print("2.Remove Item")
    print("3.Search Item")
    print("4.Calculate Total Bill")
    print("5.Display Duplicate Items")
    print("6.Show Cart")
    print("7.Exit")

    choice = int(input("Enter choice: "))

    if choice == 1:
        item = input("Enter item name: ")
        price = float(input("Enter item price: "))

        cart.append(item)
        prices.append(price)

        print("Item added successfully.")

    elif choice == 2:
        item = input("Enter item to remove: ")
```

```
        if item in cart:
            index = cart.index(item)
            cart.pop(index)
            prices.pop(index)
            print("Item removed.")
        else:
            print("Item not found.")

    elif choice == 3:
        item = input("Enter item to search: ")

        if item in cart:
            print(item, "found in cart.")
        else:
            print(item, "not found.")

    elif choice == 4:
        total = sum(prices)
        print("Total Bill =", total)

    elif choice == 5:
        duplicates = []

        for item in cart:
            if cart.count(item) > 1 and item not in duplicates:
                duplicates.append(item)

        print("Duplicate Items:", duplicates)
```

```python
    elif choice == 6:
        print("Cart Items:", cart)

    elif choice == 7:
        print("Exiting...")
        break

    else:
        print("Invalid Choice")
```

```
1.Add Item
2.Remove Item
3.Search Item
4.Calculate Total Bill
5.Display Duplicate Items
6.Show Cart
7.Exit
Enter choice:  2
Enter item to remove:  1
Item not found.

1.Add Item
2.Remove Item
3.Search Item
4.Calculate Total Bill
5.Display Duplicate Items
6.Show Cart
7.Exit
Enter choice:  7
Exiting...
```